

This exercise sheet provides some training material for the optional tutorial.

1 Proof patterns

Exercise 1

This is Exercise 1.10 in M. Joshi: Proof Patterns, Springer 2015 [2]. A statement $\mathcal{P}(k)$ is defined for all $k \in \mathbb{Z}$. We know that $\mathcal{P}(\ell)$ is true for some $\ell \in \mathbb{Z}$. Moreover, we know that for all $n \in \mathbb{Z}$,

$$\mathcal{P}(n) \rightarrow \mathcal{P}(n+1) \quad \text{as well as} \quad \mathcal{P}(n) \rightarrow \mathcal{P}(n-1).$$

is true. Prove that $\mathcal{P}(n)$ holds for all $n \in \mathbb{Z}$.

Solution 1

Suppose the induction scheme does not hold. Then $\mathcal{P}(\ell)$, $\forall n \in \mathbb{Z} (\mathcal{P}(n) \rightarrow \mathcal{P}(n+1))$, $\forall n \in \mathbb{Z} (\mathcal{P}(n) \rightarrow \mathcal{P}(n-1))$ are all true, but $\forall n \in \mathbb{Z} \mathcal{P}(n)$ is false. This implies that $\mathcal{P}(c)$ is false for some $c \in \mathbb{Z}$. Let $S^\circ = \{n \circ \ell \mid \mathcal{P}(n) \text{ is false}\}$ for $\circ \in \{>, <\}$. Observe that at least one of $S^<$, $S^>$ is non-empty.

Case 1: $S^>$ is non-empty. Then there is a *smallest* $c > \ell$ with $\mathcal{P}(c)$ false. This implies that $\mathcal{P}(c-1)$ is true. Moreover, $\mathcal{P}(n) \rightarrow \mathcal{P}(n+1)$ is true for all $n \in \mathbb{Z}$ by assumption and therefore $\mathcal{P}(c-1) \rightarrow \mathcal{P}(c)$ is true. But then $\mathcal{P}(c)$ is true resulting in a contradiction.

Case 2: $S^<$ is non-empty. Then there is a *greatest* $c < \ell$ with $\mathcal{P}(c)$ false. This implies that $\mathcal{P}(c+1)$ is true. Moreover, $\mathcal{P}(n) \rightarrow \mathcal{P}(n-1)$ is true for all $n \in \mathbb{Z}$ by assumption and therefore $\mathcal{P}(c+1) \rightarrow \mathcal{P}(c)$ is true. But then $\mathcal{P}(c)$ is true resulting in a contradiction.

We conclude that $\mathcal{P}(n)$ holds for all $n \in \mathbb{Z}$ and therefore the induction scheme holds.

Exercise 2

What is wrong with the following proof of the statement: *for any positive real x and any natural number n , $x^n = 1$ holds?*

Let $\mathcal{P}(n)$ denote $x^n = 1$. The proof is by mathematical induction on n .

Base case. For $n = 0$, $\mathcal{P}(n)$ is true because $x^0 = 1$.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq 0$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. We derive:

$$x^{n+1} = \frac{x^n \cdot x^n}{x^{n-1}}$$

$$\begin{aligned}
&= \frac{1 \cdot 1}{1} && \text{(from above by the induction hypothesis)} \\
&= 1.
\end{aligned}$$

Hence, $\mathcal{P}(n+1)$ is true. □

Solution 2

The proof is incorrect because n can be 0 and x^{-1} has not been shown to be 1 in the base case(s)!

2 Data structures

Exercise 3

Prove the following: *If a tree has $n \geq 1$ vertices, then it has $n - 1$ edges.*

Solution 3

Let $\mathcal{P}(n)$ denote the statement that any tree with n vertices has $n - 1$ edges. The proof is by strong (or complete) induction on n .

Base case: $n = 1$. A tree with one vertex contains zero edges. Therefore $\mathcal{P}(1)$ holds.

Induction hypothesis. Let $n \geq 1$ and assume that, for all $1 \leq k \leq n$, $\mathcal{P}(k)$ is true.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. Take two arbitrary nonempty trees T_1 and T_2 with $n_1 \geq 1$ and $n_2 \geq 1$ vertices such that $n_1 + n_2 = n + 1$. Then $n_1, n_2 \leq n$ holds. By the induction hypothesis, these trees have $n_1 - 1$ and $n_2 - 1$ edges. Select an arbitrary vertex in one of the trees, say in T_1 and add the other tree, T_2 , as a sub-tree (by adding an additional edge). Then the resulting tree has $n + 1$ vertices by construction and $n_1 - 1 + n_2 - 1 + 1 = n$ edges. Hence, $\mathcal{P}(n+1)$ is true. □

Remark: If we have an *inductive definition* of the set of all trees, we can alternatively use structural induction.

Exercise 4

We define the set L of all lists as follows:

$$L ::= nil \mid (c : L)$$

The symbol *nil* denotes the empty list containing no element. We define the function *append* by $append(nil, y) = y$ and $append((c : x), y) = (c : append(x, y))$. Show that, for all lists ℓ , $append(\ell, nil) = \ell$ holds.

Solution 4

Let $\mathcal{P}(\ell)$ denote the statement $\text{append}(\ell, \text{nil}) = \ell$. The proof is by structural induction on the definition of lists.

Base case: $\ell = \text{nil}$. Then $\mathcal{P}(\ell)$ is $\text{append}(\text{nil}, \text{nil}) = \text{nil}$, which is true due to the first item of the definition of append .

Induction hypothesis. Let ℓ be an arbitrary list and assume that $\mathcal{P}(\ell)$ is true.

Induction step. We want to show that $\mathcal{P}(c : \ell)$ is true. The induction hypothesis provides $\text{append}(\ell, \text{nil}) = \ell$. We extend both sides with c , resulting in

$$(c : \text{append}(\ell, \text{nil})) = (c : \ell).$$

Using the second item of the definition of append together with the transitivity of $=$, we get $\text{append}(c : \ell, \text{nil}) = (c : \ell)$. Hence, $\mathcal{P}(c : \ell)$ is true. $\square$

Exercise 5

For sets $A_1, A_2, \dots$ and $B_1, B_2, \dots$, define $B_1 = A_1$ and $B_n = A_n \setminus (\bigcup_{i=1}^{n-1} A_i)$ for $n > 1$.

Prove: $\bigcup_{i=1}^n B_i = \bigcup_{i=1}^n A_i$.

Solution 5

We first show that for any sets A and B , $(A \setminus B) \cup B = A \cup B$ holds. From exercise sheet 1, we know that $A \setminus B = A \cap \overline{B}$. Then we have:

$$\begin{aligned} (A \setminus B) \cup B &= (A \cap \overline{B}) \cup B \\ &= (A \cup B) \cap (B \cup \overline{B}) && \text{by distributivity} \\ &= (A \cup B) \cap U && U \text{ is the universal set} \\ &= (A \cup B). \end{aligned}$$

We denote by $\mathcal{P}(n)$ the statement $\bigcup_{i=1}^n B_i = \bigcup_{i=1}^n A_i$. The proof is by mathematical induction on n .

Base case: $\mathcal{P}(1)$ is true because $\bigcup_{i=1}^1 B_i = B_1 = \bigcup_{i=1}^1 A_i$.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq 1$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. By the induction hypothesis, $\mathcal{P}(n)$ is true. Then we derive

$$\begin{aligned} \bigcup_{i=1}^n B_i &= \bigcup_{i=1}^n A_i \quad \text{iff} \\ B_{n+1} \cup \bigcup_{i=1}^n B_i &= B_{n+1} \cup \bigcup_{i=1}^n A_i \quad \text{iff} \\ \bigcup_{i=1}^{n+1} B_i &= (A_{n+1} \setminus \bigcup_{i=1}^n A_i) \cup \bigcup_{i=1}^n A_i \quad \text{iff} \end{aligned}$$

$$\begin{aligned}\bigcup_{i=1}^{n+1} B_i &= A_{n+1} \cup \bigcup_{i=1}^n A_i \quad (\text{by the above lemma}) \text{ iff} \\ \bigcup_{i=1}^{n+1} B_i &= \bigcup_{i=1}^{n+1} A_i\end{aligned}$$

Hence, $\mathcal{P}(n+1)$ is true. □

3 Numbers, sums, etc.

Exercise 6

Show that any natural number above 11 can be written as a sum of 4's and 5's.

Solution 6

Let $\mathcal{P}(n)$ denote the statement that the natural number n can be written as a sum of 4's and 5's. The proof is by mathematical induction on n , starting with $n_0 = 12$.

Base cases. Since $12 = 4 + 4 + 4$, $\mathcal{P}(12)$ is true.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq 12$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. Take n and obtain the sum for n by the induction hypothesis. We distinguish two cases.

Case 1: The sum for n contains a 4. Then replace a 4 by a 5.

Case 2: The sum for n does not contain any 4. Since $n \geq 12$, this sum has at least three 5's. We simply replace the three 5 by four 4.

In both cases, we get from the sum for n a sum for $n+1$. Hence, $\mathcal{P}(n+1)$ is true. □

Exercise 7

Show that, for all positive integers n , $x^n - y^n$ is divisible by $x - y$.

Solution 7

Let $\mathcal{P}(n)$ denote the statement that $x^n - y^n$ is divisible by $x - y$. The proof is by mathematical induction on n .

Base case. $\mathcal{P}(1)$ is true, because $x^1 - y^1 = x - y$ is divisible by $x - y$.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq 1$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. Then

$$x^{n+1} - y^{n+1} = x^n(x - y) + y(x^n - y^n),$$

the first summand is divisible by $x - y$, and the second summand contains $x^n - y^n$ as a multiplicative factor. By the induction hypothesis, $x^n - y^n$ is divisible by $x - y$. Consequently, since both summands are divisible by $x - y$, so is the sum. Hence, $\mathcal{P}(n+1)$ is true. $\square$

Exercise 8

Show that for any $n \in \mathbb{N}_0$, the number $\ell_n = 4^{n+2} + 5^{2n+1}$ is divisible by 21.

Solution 8

Let $\mathcal{P}(n)$ denote the statement ℓ_n is divisible by 21.

Base case: $n = 0$. Then $4^{0+2} + 5^{2 \times 0 + 1} = 21$ is divisible by 21. Therefore $\mathcal{P}(0)$ holds.

Induction hypothesis. Assume that $\mathcal{P}(n)$ is true for some $n \geq 0$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. Then $\ell_{n+1} = 4^{n+3} + 5^{2n+3} = 4 \times 4^{n+2} + 5^2 \times 5^{2n+1} = 4 \times (4^{n+2} + 5^{2n+1}) + 21 \times 5^{2n+1}$. The former summand is divisible by 21 by the induction hypothesis, the latter summand is a multiple of 21 and therefore divisible by it. Hence, $\mathcal{P}(n+1)$ is true. $\square$

Exercise 9

Recall from former lectures the definition of *Fibonacci numbers*: $F(0) = 0$, $F(1) = 1$, and for all integers $k \geq 0$, $F(k+2) = F(k+1) + F(k)$. Show that, for all integers $n \geq 0$, $F(n) < 2^n$ holds.

Solution 9

Let n be any integer and let $\mathcal{P}(n)$ denote $F(n) < 2^n$. The proof is by mathematical induction on n , starting with two base cases for 0 and 1.

Base cases.

1. For $n = 0$, $\mathcal{P}(n)$ is true because $F(0) = 0 < 1 = 2^0$.
2. For $n = 1$, $\mathcal{P}(n)$ is true because $F(1) = 1 < 2 = 2^1$.

Induction hypothesis. Assume $\mathcal{P}(n)$ and $\mathcal{P}(n+1)$ are true for some $n \geq 0$.

Induction step. We want to show that $\mathcal{P}(n+2)$ is true. We derive:

$$\begin{array}{llll} \text{I:} & \mathcal{P}(n): & F(n) < 2^n & \text{by the induction hypothesis} \\ \text{II:} & \mathcal{P}(n+1): & F(n+1) < 2^{n+1} & \text{by the induction hypothesis} \\ \text{I + II:} & & F(n) + F(n+1) < 2^n + 2^{n+1}. & \end{array}$$

Since $F(n) + F(n+1)$ is $F(n+2)$ and $2^n + 2^{n+1} < 2^{n+1} + 2^{n+1} = 2^{n+2}$, $F(n+2) < 2^{n+2}$ follows by the transitivity of $<$. Hence, $\mathcal{P}(n+2)$ is true. $\square$

Exercise 10

Recall from former lectures the definition of *Fibonacci numbers*: $F(0) = 0$, $F(1) = 1$, and for all integers $k \geq 0$, $F(k+2) = F(k+1) + F(k)$. Show that, for all integers $n \geq 0$, $\sum_{i=0}^{n-1} F(i) < F(n+1)$ holds.

Solution 10

Let n be any non-negative integer and let $\mathcal{P}(n)$ denote $\sum_{i=0}^{n-1} F(i) < F(n+1)$. The proof is by mathematical induction on n .

Base cases. For $n = 0$, $\mathcal{P}(n)$ is true because $\sum_{i=0}^{n-1} F(i) = 0 < F(1)$.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq 0$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. We derive: $\sum_{i=0}^n F(i) = (\sum_{i=0}^{n-1} F(i)) + F(n) <_{\text{IH}} F(n+1) + F(n) = F(n+2)$. Hence, $\mathcal{P}(n+1)$ is true. $\square$

Exercise 11

Recall from former lectures the definition of *Fibonacci numbers*: $F(0) = 0$, $F(1) = 1$, and for all integers $k \geq 0$, $F(k+2) = F(k+1) + F(k)$. Show that, for all integers $n \geq 3$, $\alpha^{n-2} < F(n)$ holds, where $\alpha = (1 + \sqrt{5})/2$.

Remark: α is called the *golden ratio* and it can be proved that $\lim_{n \rightarrow \infty} F_{n+1}/F_n = \alpha$.

Solution 11

The proof has been taken from Lucia Moura's lecture slides "Induction and Recursion" (from the course on *Discrete Structures* given at the University of Ottawa).

Let $\mathcal{P}(n)$ denote the statement $\alpha^{n-2} < F(n)$. The proof is by mathematical induction on n starting with two base cases for 3 and 4.

Base cases. For $n = 3$, $\mathcal{P}(n)$ is true because $\alpha = (1 + \sqrt{5})/2 < (1 + 3)/2 = 2 = F(3)$. For $n = 4$, we have $\alpha^2 = (1 + \sqrt{5})^2/4 = (3 + \sqrt{5})/2 < 3 = F(4)$.

Induction hypothesis. Assume $\mathcal{P}(n)$ and $\mathcal{P}(n+1)$ are true for some $n \geq 3$. We want to show that $\mathcal{P}(n+2)$ is true. Then we derive:

$$\begin{aligned} F(n+2) &= F(n+1) + F(n) && \text{by the definition of the Fibonacci sequence} \\ &> \alpha^{n-1} + \alpha^{n-2} && \text{by the induction hypothesis} \\ &= \alpha^{n-2}(\alpha + 1) = \alpha^{n-2}((1 + \sqrt{5})/2 + 1) = \alpha^{n-2}((3 + \sqrt{5})/2) \\ &= \alpha^{n-2} \cdot \alpha^2 = \alpha^n. \end{aligned}$$

Hence, $\mathcal{P}(n+2)$ is true. $\square$

Exercise 12

Let $\mathcal{P}(n)$ denote the statement $n! > 2^n$. Show that there is a smallest natural number n_0 such that $\mathcal{P}(n)$ holds for all natural numbers $n \geq n_0$.

Solution 12

For $n_0 = 1, 2, 3$, $\mathcal{P}(n)$ is false and for $n_0 = 4$ it is true. The proof is by mathematical induction on n starting with $n_0 = 4$.

Base case: $n_0 = 4$. $\mathcal{P}(n_0)$ holds because $4! > 2^4$.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq n_0$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. By the induction hypothesis, $\mathcal{P}(n)$ is true. Then we derive

$$\begin{aligned} n! &> 2^n \quad \text{iff} \\ (n+1)n! &> (n+1)2^n \quad \text{iff} \\ (n+1)! &> (n+1)2^n. \end{aligned}$$

With $n \geq 4$, $(n+1)2^n > 2^{n+1}$ holds. Then $(n+1)! > 2^{n+1}$ follows by transitivity of $>$. Hence, $\mathcal{P}(n+1)$ is true. $\square$

Exercise 13

Let us define the *Fermat* numbers $F_n = 2^{(2^n)} + 1$ for all natural numbers $n \geq 0$. Prove that for all $n \geq 1$, $F_n = (F_0 \cdot F_1 \cdots F_{n-1}) + 2$.

Solution 13

Let $\mathcal{P}(n)$ denote the statement $F_n = (F_0 \cdot F_1 \cdots F_{n-1}) + 2$. The proof is by mathematical induction on n .

Base case: $n = 1$. Observe that $F_0 = 3$ and $F_1 = 5$ by the definition of the Fermat numbers. Then $F_1 = F_0 + 2$ and $\mathcal{P}(1)$ holds.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq 1$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true.

$$\begin{aligned} F_{n+1} &= (F_0 \cdot F_1 \cdots F_n) + 2 \\ &= ((F_n - 2) \cdot F_n) + 2 \quad (\text{by the induction hypothesis}) \\ &= F_n^2 - 2F_n + 2 \\ &= (2^{(2^n)} + 1)^2 - 2(2^{(2^n)} + 1) + 2 \\ &= (2^{(2^n)})^2 + 2 \cdot 2^{(2^n)} + 1 - 2 \cdot 2^{(2^n)} \\ &= (2^{(2^n)})^2 + 1 \\ &= 2^{(2^{n+1})} + 1 \end{aligned}$$

Hence, $\mathcal{P}(n+1)$ is true. $\square$

Exercise 14

From M. Sipser: Introduction to the theory of computation, third edition.

Let $S(n) = 1 + 2 + \dots + n$ be the sum of the first n natural numbers and let $C(n) = 1^3 + 2^3 + \dots + n^3$ be the sum of the first n cubes. Prove the following equalities by induction on n , to arrive at the curious conclusion that $C(n) = (S(n))^2$ for every n .

1. $S(n) = \frac{1}{2}n(n+1)$.
2. $C(n) = \frac{1}{4}(n^4 + 2n^3 + n^2) = \frac{1}{4}n^2(n+1)^2$.

Solution 14

We show 1. and 2. by mathematical induction on n .

Base case: $n = 1$. Then

$$\begin{aligned} S(1) &= \frac{1}{2}n(n+1) = \frac{1}{2}1(1+1) = 1 \\ C(1) &= \frac{1}{4}n^2(n+1)^2 = \frac{1}{4}1^2(1+1)^2 = 1 \end{aligned}$$

Induction hypothesis. Let $n \geq 1$ be some natural number and assume that the following hold: $S(n) = \frac{1}{2}n(n+1)$ and $C(n) = \frac{1}{4}n^2(n+1)^2$.

Induction step. We want to show that $S(n+1) = \frac{1}{2}(n+1)(n+2)$ and $C(n+1) = \frac{1}{4}(n+1)^2(n+2)^2$ hold.

$$\begin{aligned} S(n+1) &= S(n) + (n+1) = \frac{1}{2}n(n+1) + (n+1) = \frac{1}{2}(n+1)(n+2) \\ C(n+1) &= C(n) + (n+1)^3 = \frac{1}{4}n^2(n+1)^2 + (n+1)^3 \\ &= \frac{1}{4}n^2(n+1)^2 + (n+1)(n+1)^2 = \frac{1}{4}(n+1)^2(n^2 + 4(n+1)) \\ &= \frac{1}{4}(n+1)^2(n^2 + 4n + 4) = \frac{1}{4}(n+1)^2(n+2)^2 \end{aligned}$$

4 Logic and induction

Exercise 15

Let I, I' be interpretations with domains $\mathcal{U}, \mathcal{U}'$. We say that I is *isomorphic* to I' if there exists a bijection $\chi : \mathcal{U} \rightarrow \mathcal{U}'$ such that the following conditions are fulfilled:

1. $\chi(I(c)) = I'(c)$ for every constant symbol c .
2. $\chi(I(f)(p_1, \dots, p_n)) = I'(f)(\chi(p_1), \dots, \chi(p_n))$ for every n -ary function symbol f ($n > 0$) and all $p_1, \dots, p_n \in \mathcal{U}$.

χ is called *isomorphism* between I and I' .

Now let χ be an isomorphism between I and I' . Prove that for all closed (i.e., variable-free) terms t we have $I'(t) = \chi(I(t))$.

Hint: Prove the statement by structural induction on t .

Solution 15

Let $\mathcal{P}(t)$ denote the statement $I'(t) = \chi(I(t))$ for a closed term t . The proof is by structural induction on the definition of terms. Observe that we have to deal with a single base case, since t cannot be a variable by assumption.

Base case: t is an arbitrary constant, say c . We obtain $I'(t) = I'(c) = \chi(I(c)) = \chi(I(t))$ by item 1. Then $\mathcal{P}(t)$ is true.

Induction hypothesis. Let $t_1, \dots, t_n$ be arbitrary closed term and assume that $\mathcal{P}(t_1) \wedge \dots \wedge \mathcal{P}(t_n)$ is true.

Induction step. We want to show that $\mathcal{P}(f(t_1, \dots, t_n))$ is true, where f is an arbitrary function symbol of arity n . We obtain $I'(f(t_1, \dots, t_n)) = I'(f)(I'(t_1), \dots, I'(t_n))$ by the definition of the evaluation of a term. Applying the induction hypothesis, we obtain $I'(f(t_1, \dots, t_n)) = I'(f)(\chi(I(t_1)), \dots, \chi(I(t_n)))$. From the latter together with item 2, we obtain $I'(f(t_1, \dots, t_n)) = \chi(I(f)(I(t_1), \dots, I(t_n)))$ by symmetry and transitivity of $=$. Therefore $\mathcal{P}(f(t_1, \dots, t_n))$ is true. $\square$

Exercise 16 Equivalent Replacement Theorem

Prove the following statement:

Let φ , ψ_1 , and ψ_2 be propositional formulas. If $\psi_1 \equiv \psi_2$, then $\varphi[\psi_1] \equiv \varphi[\psi_2]$.

Solution 16

from solutions of FMInf ex sheet on SAT

5 Binomial coefficients and induction

We consider properties of binomial coefficients in the following. Here are some basic facts and definitions about them.

1. We write binomial coefficients in the form $\binom{a}{b}$, where a and b are non-negative integers.
2. For convenience we allow that b is an integer.
3. $\binom{a}{b}$ is calculated as follows: (i) If $a < b$ or $b < 0$ then $\binom{a}{b} = 0$ by definition. Otherwise $\binom{a}{b} = \frac{a!}{b!(a-b)!}$.
4. $\binom{a}{b} = \binom{a}{a-b}$ and $\binom{a}{0} = \binom{a}{a} = 1$.

5. We will often use that $\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$ for any integer n with $n \geq 0$ and integer r with $0 \leq r \leq n$.

Exercise 17

Show for all non-negative integers n that the following holds:

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$

Solution 17

Let $\mathcal{P}(n)$ denote the statement

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$

The proof is by mathematical induction on n .

Base case: $n = 0$. Then $\binom{0}{0} = 1 = 2^0$. Hence, $\mathcal{P}(0)$ is true.

Induction hypothesis. Let $n \geq 0$ be some integer and assume that $\mathcal{P}(n)$ is true.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true.

$$\begin{aligned} & \binom{n+1}{0} + \binom{n+1}{1} + \cdots + \binom{n+1}{n} + \binom{n+1}{n+1} \\ &= 1 + \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n-1} + \binom{n}{n} + 1 \\ &= \binom{n}{0} + \cdots + \binom{n}{n-1} + \binom{n}{n} + \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} =_{\text{IH}} 2^n + 2^n = 2^{n+1} \end{aligned}$$

Hence $\mathcal{P}(n+1)$ is true. □

Exercise 18

Show for all positive integers n that the following holds:

$$\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}.$$

Solution 18

Let $\mathcal{P}(n)$ denote the statement $\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}$. The proof is by mathematical induction on n .

Base case: $n = 1$. Then $\mathcal{P}(1)$ is true because

$$\sum_{k=1}^n k \binom{n}{k} = 1 \binom{1}{1} = 1 = 2^0 = n2^{n-1}.$$

Induction hypothesis. Let $n \geq 1$ be some integer and assume that $\mathcal{P}(n)$ is true.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true.

$$\begin{aligned} \sum_{k=1}^{n+1} k \binom{n+1}{k} &= (n+1) + \sum_{k=1}^n k \binom{n+1}{k} = (n+1) + \sum_{k=1}^n k \binom{n}{k} + \sum_{k=1}^n k \binom{n}{k-1} \\ &=_{\text{IH}} (n+1) + n2^{n-1} + \sum_{k=1}^n k \binom{n}{k-1} \\ &= (n+1) + n2^{n-1} + \sum_{k=0}^{n-1} (k+1) \binom{n}{k} \\ &= n2^{n-1} + \sum_{k=0}^n (k+1) \binom{n}{k} = n2^{n-1} + 1 + \sum_{k=1}^n (k+1) \binom{n}{k} \\ &= n2^{n-1} + 1 + \sum_{k=1}^n k \binom{n}{k} + \sum_{k=1}^n \binom{n}{k} = n2^{n-1} + \sum_{k=1}^n k \binom{n}{k} + \sum_{k=0}^n \binom{n}{k} \\ &=_{\text{IH}} n2^{n-1} + n2^{n-1} + \sum_{k=0}^n \binom{n}{k} = n2^n + 2^n = (n+1)2^n \end{aligned}$$

Hence $\mathcal{P}(n+1)$ is true. □

Exercise 19

Show for all non-negative integers n and m that the following holds:

$$\sum_{k=0}^n k \binom{m+k}{m} = n \binom{m+n+1}{m+1} - \binom{m+n+1}{m+2}.$$

Solution 19

Let us fix an arbitrary non-negative integer m . Let $\mathcal{P}(n)$ denote the statement

$$\sum_{k=0}^n k \binom{m+k}{m} = n \binom{m+n+1}{m+1} - \binom{m+n+1}{m+2}.$$

The proof is by mathematical induction on n .

Base case: $n = 0$. Then $\sum_{k=0}^0 k \binom{m+k}{m} = 0 = 0 \binom{m+0+1}{m+1} - \binom{m+0+1}{m+2}$. Hence $\mathcal{P}(0)$ is true.

Induction hypothesis. Let $n \geq 1$ be some integer and assume that $\mathcal{P}(n)$ is true.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true.

$$\begin{aligned}
\sum_{k=0}^{n+1} k \binom{m+k}{m} &= \sum_{k=0}^n k \binom{m+k}{m} + (n+1) \binom{m+n+1}{m} \\
&=_{\text{IH}} n \binom{m+n+1}{m+1} - \binom{m+n+1}{m+2} + (n+1) \binom{m+n+1}{m} \\
&= (n+1) \left[\binom{m+n+1}{m} + \binom{m+n+1}{m+1} \right] - \binom{m+n+1}{m+1} - \binom{m+n+1}{m+2} \\
&= (n+1) \binom{m+n+1+1}{m+1} - \binom{m+n+1+1}{m+2}
\end{aligned}$$

Hence $\mathcal{P}(n+1)$ is true. □

Exercise 20

Show for all integers $n > 0$ that the alternating sum of the binomial coefficients is zero, i.e., show that

$$\sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots \pm \binom{n}{n} = 0.$$

What can you say about the sub-sum of all binomial coefficients $\binom{n}{\ell}$ with $\ell \leq n$ and ℓ even and the similar sub-sum with ℓ odd?

Solution 20

Let $\mathcal{P}(n)$ denote the statement $\sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} = 0$. The proof is by mathematical induction on n .

Base case: $n = 1$. Then $\binom{1}{0} - \binom{1}{1} = 1 - 1 = 0$. Hence $\mathcal{P}(1)$ is true.

Induction hypothesis. Let $n \geq 1$ be some integer and assume that $\mathcal{P}(n)$ is true.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true.

$$\begin{aligned}
\sum_{\ell=0}^{n+1} (-1)^\ell \binom{n+1}{\ell} &= (-1)^0 + (-1)^{n+1} + \sum_{\ell=1}^n (-1)^\ell \left[\binom{n}{\ell-1} + \binom{n}{\ell} \right] \\
&= (-1)^{n+1} + \sum_{\ell=1}^n (-1)^\ell \binom{n}{\ell-1} + (-1)^0 + \sum_{\ell=1}^n (-1)^\ell \binom{n}{\ell} \\
&= (-1)^{n+1} + \sum_{\ell=1}^n (-1)^\ell \binom{n}{\ell-1} + \sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} \\
&=_{\text{IH}} (-1)^{n+1} + \sum_{\ell=1}^n (-1)^\ell \binom{n}{\ell-1} = (-1)^{n+1} + \sum_{\ell=0}^{n-1} (-1)^{\ell+1} \binom{n}{\ell}
\end{aligned}$$

$$= \sum_{\ell=0}^n (-1)^{\ell+1} \binom{n}{\ell} = (-1) \sum_{\ell=0}^n (-1)^{\ell} \binom{n}{\ell} =_{\text{IH}} 0$$

Hence $\mathcal{P}(n+1)$ is true. The value of the sub-sums are identical but they have different signs in the total sum and are therefore cancel out. $\square$

Exercise 21

Show for all integers $n \geq 1$ and all $\ell \geq 1$ that

$$\sum_{i=0}^{\ell} \binom{n+i}{i} = \binom{n}{0} + \binom{n+1}{1} + \cdots + \binom{n+\ell}{\ell} = \binom{n+\ell+1}{\ell}.$$

Hint: Use mathematical induction on ℓ .

Solution 21

Let $\mathcal{P}(\ell)$ denote the statement $\sum_{i=0}^{\ell} \binom{n+i}{i} = \binom{n+\ell+1}{\ell}$ for all integers $n \geq 1$. The proof is by mathematical induction on ℓ .

Base case: $\ell = 1$. Fix an arbitrary n with $n \geq 1$. Then

$$\binom{n}{0} + \binom{n+1}{1} = 1 + (n+1) = \binom{n+1+1}{1} = \binom{n+\ell+1}{\ell}$$

We have verified that $\mathcal{P}(1)$ is true.

Induction hypothesis. Let $\ell \geq 1$ be some integer and assume that $\mathcal{P}(\ell)$ is true.

Induction step. We want to show that $\mathcal{P}(\ell+1)$ is true.

Fix an arbitrary integer n with $n \geq 1$.

$$\begin{aligned} \sum_{k=0}^{\ell+1} \binom{n+k}{k} &= \sum_{k=0}^{\ell} \binom{n+k}{k} + \binom{n+\ell+1}{\ell+1} =_{\text{IH}} \binom{n+\ell+1}{\ell} + \binom{n+\ell+1}{\ell+1} \\ &= \binom{n+(\ell+1)+1}{\ell+1} \end{aligned}$$

Hence $\mathcal{P}(\ell+1)$ is true. $\square$

Exercise 22

Show that, for all non-negative integers m , k , and n , the following holds:

$$\binom{m+n}{k} = \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i}$$

What can you say about the value of $\binom{2n}{n}$?

Hint: Use mathematical induction on m .

Solution 22

Let $\mathcal{P}(m)$ denote the statement that

$$\binom{m+n}{k} = \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i}$$

is true for all non-negative integers k and n . The proof is by mathematical induction on m .

Base case: $m = 0$. Choose two arbitrary non-negative integers k and n . Then, with $\binom{m}{i} = 0$ if $i > m$,

$$\sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} = \sum_{i=0}^k \binom{0}{i} \binom{n}{k-i} = \binom{0}{0} \binom{n}{k} = \binom{n}{k} = \binom{m+n}{k}$$

We have verified that $\mathcal{P}(0)$ is true.

Induction hypothesis. Let $m \geq 0$ be some integer and assume that $\mathcal{P}(m)$ is true.

Induction step. We want to show that $\mathcal{P}(m+1)$ is true.

$$\begin{aligned} \sum_{i=0}^k \binom{m+1}{i} \binom{n}{k-i} &= \sum_{i=1}^k \binom{m+1}{i} \binom{n}{k-i} + \binom{n}{k} \\ &= \sum_{i=1}^k \binom{m}{i} \binom{n}{k-i} + \sum_{i=1}^k \binom{m}{i-1} \binom{n}{k-i} + \binom{n}{k} \\ &= \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} + \sum_{i=1}^k \binom{m}{i-1} \binom{n}{k-i} \\ &= \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} + \sum_{i=0}^{k-1} \binom{m}{i} \binom{n}{k-1-i} \\ &=_{\text{IH}} \binom{m+n}{k} + \binom{m+n}{k-1} = \binom{m+1+n}{k} \end{aligned}$$

Hence $\mathcal{P}(m+1)$ is true. □

Remark: We apply the induction hypothesis twice, once to a binomial coefficient where k is involved and once to one with $k-1$. This is the reason why we need the (strong) induction hypothesis with the universal quantifiers inside $\mathcal{P}$.

The value of $\binom{2n}{n}$ follows from the above by setting $m = n = k$. Then

$$\binom{2n}{n} = \binom{n+n}{n} = \sum_{i=0}^n \binom{n}{i} \binom{n}{n-i} = \sum_{i=0}^n \binom{n}{i}^2$$

Exercise 23

For positive integers m , n , and p , show that the following holds:

$$\binom{m+n}{m+p} = \sum_{k=0}^k \binom{m}{k} \binom{n}{p+k}$$

Solution 23

Let $\mathcal{P}(m)$ denote the statement that

$$\binom{m+n}{m+p} = \sum_{k=0}^k \binom{m}{k} \binom{n}{p+k}$$

is true for all positive integers n and p . The proof is by mathematical induction on m .

Base case: $m = 1$. Choose two arbitrary positive integers n and p . Then

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{p+k} = \binom{1}{0} \binom{n}{p} + \binom{1}{1} \binom{n}{p+1} = \binom{n}{p} + \binom{n}{p+1} = \binom{n+1}{p+1}$$

We have verified that $\mathcal{P}(0)$ is true.

Induction hypothesis. Let $m \geq 1$ be some integer and assume that $\mathcal{P}(m)$ is true.

Induction step. We want to show that $\mathcal{P}(m+1)$ is true

Choose two arbitrary positive integers n and p . Then

$$\begin{aligned} \sum_{k=0}^{m+1} \binom{m+1}{k} \binom{n}{p+k} &= \sum_{k=1}^{m+1} \binom{m+1}{k} \binom{n}{p+k} + \binom{m+1}{0} \binom{n}{p} \\ &= \sum_{k=1}^{m+1} \left[\binom{m}{k} + \binom{m}{k-1} \right] \binom{n}{p+k} + \binom{n}{p} \\ &= \sum_{k=1}^{m+1} \binom{m}{k} \binom{n}{p+k} + \sum_{k=1}^{m+1} \binom{m}{k-1} \binom{n}{p+k} + \binom{n}{p} \\ &= \sum_{k=0}^{m+1} \binom{m}{k} \binom{n}{p+k} + \sum_{k=1}^{m+1} \binom{m}{k-1} \binom{n}{p+k} \\ &= \sum_{k=0}^m \binom{m}{k} \binom{n}{p+k} + \sum_{k=0}^m \binom{m}{k} \binom{n}{p+k+1} \\ &=_{\text{IH}} \binom{m+n}{m+p} + \binom{m+n}{m+p+1} = \binom{m+1+n}{m+1+p} \end{aligned}$$

Hence $\mathcal{P}(m+1)$ is true. □

Exercise 24

Let $a, b \in \mathbb{R}$, $n \in \mathbb{N}$. Then $(a + b)^n = \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i$.

Solution 24

We first show that $\binom{n}{i} + \binom{n}{i-1} = \binom{n+1}{i}$ for $1 \leq i \leq n$.

$$\begin{aligned} \binom{n}{i} + \binom{n}{i-1} &= \frac{n!}{(n-i)!i!} + \frac{n!}{(n-i+1)!(i-1)!} \\ &= \frac{n!(n-i+1)}{(n-i+1)!i!} + \frac{n!i}{(n-i+1)!i!} \\ &= \frac{n!(n-i+1+i)}{(n-i+1)!i!} = \frac{n!(n+1)}{(n-i+1)!i!} = \frac{(n+1)!}{(n+1-i)!i!} \\ &= \binom{n+1}{i} \end{aligned}$$

Let $\mathcal{P}(n)$ denote the statement $(a+b)^n = \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i$. The proof is by mathematical induction on n .

Base case: $n = 1$. Then $(a+b)^1 = \binom{1}{0}a^1b^0 + \binom{1}{1}a^0b^1 = a+b$.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq 1$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true.

$$\begin{aligned} (a+b)^{n+1} &= (a+b)(a+b)^n \stackrel{\text{IH}}{=} (a+b) \left[\sum_{i=0}^n \binom{n}{i} a^{n-i} b^i \right] \\ &= \sum_{i=0}^n \binom{n}{i} a^{n-i+1} b^i + \sum_{i=0}^n \binom{n}{i} a^{n-i} b^{i+1} \end{aligned}$$

With $j = i+1$, we get

$$\begin{aligned} (a+b)^{n+1} &= \sum_{i=0}^n \binom{n}{i} a^{n-i+1} b^i + \sum_{j=1}^{n+1} \binom{n}{j-1} a^{n-j+1} b^j \\ &= \sum_{i=0}^n \binom{n}{i} a^{n-i+1} b^i + \sum_{i=1}^{n+1} \binom{n}{i-1} a^{n-i+1} b^i \\ &= \binom{n}{0} a^{n+1} + \left(\sum_{i=1}^n \left[\binom{n}{i} + \binom{n}{i-1} \right] a^{n-i+1} b^i \right) + \binom{n}{n} b^{n+1} \end{aligned}$$

With $\binom{n}{i} + \binom{n}{i-1} = \binom{n+1}{i}$, we have

$$(a+b)^{n+1} = \binom{n}{0} a^{n+1} + \left(\sum_{i=1}^n \binom{n+1}{i} a^{n-i+1} b^i \right) + \binom{n}{n} b^{n+1}$$

$$\begin{aligned}
&= \binom{n+1}{0} a^{n+1} + \left(\sum_{i=1}^n \binom{n+1}{i} a^{n-i+1} b^i \right) + \binom{n+1}{n+1} b^{n+1} \\
&= \sum_{i=0}^{n+1} \binom{n+1}{i} a^{n+1-i} b^i
\end{aligned}$$

Hence, $\mathcal{P}(n+1)$ is true. $\square$

Exercise 25

For all non-negative integers n and all real numbers x with $x > -1$, $(1+x)^n \geq 1+nx$ holds.

Solution 25

Let $\mathcal{P}(n)$ denote the statement that $(1+x)^n \geq 1+nx$ holds for all real numbers x with $x > -1$. The proof is by mathematical induction on n .

Base case: $n = 0$. Then $(1+x)^0 = 1 \geq 1+nx$ for any real number x . The statement $\mathcal{P}(0)$ is true.

Induction hypothesis. Let $n \geq 0$ be some integer and assume that $\mathcal{P}(n)$ is true.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true.

Choose an arbitrary real number x with $x > -1$. Take $(1+x)^n \geq 1+nx$ provided by the induction hypothesis and observe that $1+x$ is positive because $x > -1$ by assumption. Then

$$\begin{aligned}
(1+x)^n &\geq 1+nx && \text{iff} \\
(1+x)^n(1+x) &\geq (1+nx)(1+x) && \text{iff} \\
(1+x)^{n+1} &\geq 1+x+nx+nx^2 && \text{iff} \\
(1+x)^{n+1} &\geq 1+(n+1)x+nx^2
\end{aligned}$$

Then $(1+x)^{n+1} \geq 1+(n+1)x$ holds for all real numbers x with $x > -1$ because nx^2 is non-negative. Hence $\mathcal{P}(n+1)$ is true. $\square$

6 More applications

Exercise 26

Suppose we have an operation (denoted by $\prime$) which is applied to symbols. The operation $\prime$ satisfies the following three axioms:

$$\begin{aligned}
(u+v)' &= u' + v' && (A+)' \\
(uv)' &= uv' + u'v && (A\cdot)' \\
(cu)' &= cu' && (Ac)'
\end{aligned}$$

Define $w^{(k)}$ inductively as follows:

B: $w^{(0)} = w$

S: Suppose $w^{(n)}$ is already defined for some natural number $n \geq 0$. Then $w^{(n+1)} = (w^{(n)})'$.

Prove Leibniz's formula $(uv)^{(n)} = \sum_{i=0}^n \binom{n}{i} u^{(i)} v^{(n-i)}$. (Recall that $\binom{n}{i} + \binom{n}{i-1} = \binom{n+1}{i}$.)

Solution 26

Let $\mathcal{P}(n)$ denote the statement $(uv)^{(n)} = \sum_{i=0}^n \binom{n}{i} u^{(i)} v^{(n-i)}$. The proof is by mathematical induction on n .

Base case: $n = 0$. Then $(uv)^{(0)} = \binom{0}{0} u^{(0)} v^{(0)} = uv$.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq 0$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. By the induction hypothesis, we know that $(uv)^{(n)} = \sum_{i=0}^n \binom{n}{i} u^{(i)} v^{(n-i)}$.

$$\begin{aligned}
(uv)^{(n+1)} &= \left((uv)^{(n)} \right)' = \left(\sum_{i=0}^n \binom{n}{i} u^{(i)} v^{(n-i)} \right)' \\
&= \sum_{i=0}^n \left(\binom{n}{i} u^{(i)} v^{(n-i)} \right)' && \text{(by } A+') \\
&= \sum_{i=0}^n \binom{n}{i} \left(u^{(i)} v^{(n-i)} \right)' && \text{(by } Ac') \\
&= \sum_{i=0}^n \binom{n}{i} \left(u^{(i)} (v^{(n-i)})' + (u^{(i)})' v^{(n-i)} \right) && \text{(by } A') \\
&= \sum_{i=0}^n \binom{n}{i} \left(u^{(i)} v^{(n-i+1)} + u^{(i+1)} v^{(n-i)} \right) \\
&= \sum_{i=0}^n \binom{n}{i} u^{(i)} v^{(n-i+1)} + \sum_{i=0}^n \binom{n}{i} u^{(i+1)} v^{(n-i)}
\end{aligned}$$

With $j = i + 1$, we get

$$\begin{aligned}
(uv)^{(n+1)} &= \sum_{i=0}^n \binom{n}{i} u^{(i)} v^{(n-i+1)} + \sum_{j=1}^{n+1} \binom{n}{j-1} u^{(j)} v^{(n-j+1)} \\
&= \sum_{i=0}^n \binom{n}{i} u^{(i)} v^{(n-i+1)} + \sum_{i=1}^{n+1} \binom{n}{i-1} u^{(i)} v^{(n-i+1)} \\
&= \binom{n}{0} u^{(0)} v^{(n+1)} + \left(\sum_{i=1}^n \left[\binom{n}{i} + \binom{n}{i-1} \right] u^{(i)} v^{(n-i+1)} \right) + \binom{n}{n} u^{(n+1)} v^{(0)}
\end{aligned}$$

With $\binom{n}{i} + \binom{n}{i-1} = \binom{n+1}{i}$, we have

$$\begin{aligned}
(uv)^{(n+1)} &= \binom{n}{0} u^{(0)} v^{(n+1)} + \left(\sum_{i=1}^n \binom{n+1}{i} u^{(i)} v^{(n-i+1)} \right) + \binom{n}{n} u^{(n+1)} v^{(0)} \\
&= \binom{n+1}{0} u^{(0)} v^{(n+1)} + \left(\sum_{i=1}^n \binom{n+1}{i} u^{(i)} v^{(n-i+1)} \right) + \binom{n+1}{n+1} u^{(n+1)} v^{(0)} \\
&= \sum_{i=0}^{n+1} \binom{n+1}{i} u^{(i)} v^{(n+1-i)}
\end{aligned}$$

Hence, $\mathcal{P}(n+1)$ is true. $\square$

Exercise 27

This exercise is taken from Hopcroft, J.E., Motwani, R. and Ullman, J.D.: *Introduction to Automata Theory, Languages, and Computation*, 3rd ed., Pearson, 2006 [1]. It requires some background from automata theory.

A non-deterministic finite automaton A is a quintuple $(Q, \Sigma, \delta, q_0, F)$, where Q is a finite set of states, Σ is a finite set of input symbols, $q_0 \in Q$ is the start state, and $F \subseteq Q$ is the set of final (or accepting) states. The transition function δ takes a state from Q and an input symbol from Σ as arguments and returns a subset of Q .

The set of all strings over Σ is denoted by Σ^* . Let x and y be two strings. Then xy is the concatenation of them. The length of a string w (i.e., the number of symbol occurrences in w) is denoted by $|w|$. The empty string ϵ is of length 0.

We denote by $\hat{\delta}$ the extension of δ to strings.

$$\hat{\delta}(q, w) = \begin{cases} \{q\} & \text{if } w = \epsilon \text{ (the empty string);} \\ \bigcup_{p \in \delta(q, x)} \delta(p, a) & \text{if } w = xa, x \in \Sigma^* \text{ and } a \in \Sigma. \end{cases}$$

The language accepted by A , $L(A)$, is $\{w \mid \hat{\delta}(q_0, w) \cap F \neq \{\}\}$.

Let $A = (\{q_0, q_1, q_2\}, \{0, 1\}, \delta, q_0, \{q_2\})$, where $\delta(q_0, 0) = \{q_0, q_1\}$, $\delta(q_0, 1) = \{q_0\}$, $\delta(q_1, 0) = \{\}$, $\delta(q_1, 1) = \{q_2\}$, and $\delta(q_2, 0) = \delta(q_2, 1) = \{\}$. Show that $L(A) = \{w \mid w \text{ ends in } 01\}$.

Hint. Use mutual induction on the following statements:

1. $\hat{\delta}(q_0, w)$ contains q_0 for every w .
2. $\hat{\delta}(q_0, w)$ contains q_1 if and only if w ends in 0.
3. $\hat{\delta}(q_0, w)$ contains q_2 if and only if w ends in 01.

Solution 27

See pages 59 and 60 *op. cit.*

Remark: *op. cit.* abbreviates *opere citato* (Latin for “in the work cited”).

Exercise 28

Consider a (generalized) chess board of size $2^n \times 2^n$, where one position is cut out. Take an L-tile made of three positions and show for all natural numbers $n \geq 1$ that the chess board can be covered using the L-tiles. Compute the number of required L-tiles.

Solution 28

Playing with some cases for small values of n , we came up with the number $\ell(n) = \sum_{k=0}^{n-1} 2^{2k}$ which we will use in the following. Let $\mathcal{P}(n)$ denote the statement “The $2^n \times 2^n$ chess board with one missing position can be covered by L-tiles and we need $\ell(n) = \sum_{k=0}^{n-1} 2^{2k}$ of them.” The proof is by mathematical induction on n .

Base case: $n = 1$. Then we have a 2×2 chess board. There are 4 possible positions which can be cut out. In each of these cases, one L-tile covers the three remaining positions. Hence, $\ell(1) = 1 = \sum_{k=0}^{n-1} 2^{2k}$.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq 1$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true.

Consider a $2^{n+1} \times 2^{n+1}$ chess board with one missing position. Decompose it into four chess boards of size $2^n \times 2^n$, one of which has a missing position. Now we arrange one L-tile in such a way that it affects all the three full boards. After putting this L-tile, there is one position missing/already covered in any of the four boards. Therefore the induction hypothesis can be applied to each of these sub-boards of size $2^n \times 2^n$. Hence all four sub-boards (and therefore the $2^{n+1} \times 2^{n+1}$ board) can be covered. Then

$$\begin{aligned}\ell(n+1) &= 4\ell(n) + 1 = 2^2 \left(\sum_{k=0}^{n-1} 2^{2k} \right) + 1 = \left(\sum_{k=0}^{n-1} 2^{2(k+1)} \right) + 1 \\ &= \left(\sum_{k=1}^n 2^{2k} \right) + 1 = \sum_{k=0}^n 2^{2k}\end{aligned}$$

Hence, $\mathcal{P}(n+1)$ is true. □

Exercise 29

You have a bag with red, yellow and blue chips. If only one chip remains in the bag, you put it out. Otherwise you remove two chips at random:

1. If one of the removed chips is red, you do not put any chips in the bag.
2. If both of the removed chips are yellow, you put one yellow chip and five blue chips in the bag.
3. If one of the chips is blue and the other is not red, you put ten red chips in the bag.

Show that any sequence of moves applied to an arbitrary bag always terminates or provide a non-terminating sequence of moves.

Solution 29

Represent the state of the bag by a triple $\langle y, b, r \rangle$ of non-negative integers such that y, b, r represents the number of yellow, blue, red chips in the bag. Let $<_3$ be the natural lexicographic extension to such triples from $\mathbb{N}_0^3$.

We prove by well-founded induction using $<_3$ that the process eventually terminates with an empty bag. Let $\mathcal{P}(S)$ denote that the process terminates for state $S = \langle y, b, r \rangle$.

Base case: Consider the minimal state $S = \langle y, b, r \rangle = \langle 0, 0, 0 \rangle$. Then the bag is empty and the process terminates immediately. Therefore $\mathcal{P}(S) = \mathcal{P}(\langle 0, 0, 0 \rangle)$ holds.

Induction hypothesis. Let S be some (non-minimal) state and suppose that $\mathcal{P}(S')$ holds for all states S' with $S' <_3 S$.

Induction step. We want to show that the process starting with $\mathcal{P}(S) = \langle y, b, r \rangle$ terminates. Take S and let us remove chips. Then we distinguish the following four possible cases.

Case 1: S is of the form $\langle 1, 0, 0 \rangle$, $\langle 0, 1, 0 \rangle$ or $\langle 0, 0, 1 \rangle$. Since we can take a single remaining chip from the bag, the resulting bag corresponds to the minimal state, which is less than S with respect to $<_3$.

Case 2: In this and the following cases, we can remove 2 chips. We apply move (a) to the state S resulting in one of the states $\langle y-1, b, r-1 \rangle$, $\langle y, b-1, r-1 \rangle$ or $\langle y, b, r-2 \rangle$. Each of the new states is less than S with respect to $<_3$.

Case 3: We apply move (b) to the state S resulting in the state $\langle y-1, b+5, r \rangle$, which is less than S with respect to $<_3$.

Case 4: We apply move (c) to the state S resulting in one of the states $\langle y-1, b-1, r+10 \rangle$ or $\langle y, b-2, r+10 \rangle$. Each of the new states is less than S with respect to $<_3$.

In each of the four cases, the induction hypothesis provides termination of the process with starting bag corresponding to the smaller state S' . Since we reach S from S' by one more move, we conclude that the process terminates for S .

Exercise 30

This exercise requires some background from linear algebra, which we take from https://en.wikipedia.org/wiki/Matrix_exponential.

A *matrix exponential* is a matrix function defined as

$$e^M = \sum_{k=0}^{\infty} \frac{1}{k!} M^k = I + M + \frac{M^2}{2!} + \dots$$

where M is an $n \times n$ (real or complex) matrix and $M^0 = I$ is the *identity* matrix. The series e^M *converges absolutely* (see <http://math.ucr.edu/~res/math138A/expmatrix.pdf> for the definition and a proof). Absolute convergence implies that manipulations on

infinite sums like the Cauchy product

$$\left(\sum_{i=0}^{\infty} A_i\right)\left(\sum_{j=0}^{\infty} B_j\right) = \left(\sum_{i,j=0}^{\infty} A_i B_j\right) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n A_k B_{n-k}\right)$$

can be safely performed.

Show the following:

1. If $AB = BA$ then $A^\ell e^B = e^B A^\ell$ and $e^A e^B = e^B e^A$ for arbitrary $n \times n$ (real or complex) matrices A and B , and $\ell \in \mathbb{N}_0$.
2. $e^{(s+t)A} = e^{sA} e^{tA}$ for an arbitrary $n \times n$ (real or complex) matrix A and (real or complex) numbers s and t . What can you say about the inverse of e^A ?

Solution 30

Let us assume that $AB = BA$ holds. We first show by strong mathematical induction on ℓ that $A^\ell e^B = e^B A^\ell$.

Base cases: $\ell = 0$. Then $A^0 e^B = I e^B = e^B I = e^B A^0$.

We explicitly state the case for $\ell = 1$, because we will use it later on. Then

$$A e^B = A \left(\sum_{k=0}^{\infty} \frac{1}{k!} B^k \right) = AI + AB + \frac{AB^2}{2!} + \dots = IA + BA + \frac{B^2 A}{2!} + \dots = e^B A.$$

Induction hypothesis. Let $\ell \geq 0$ be some natural number and assume that $A^m e^B = e^B A^m$ holds for all $m = 0, \dots, \ell$.

Induction step. We want to show that $A^{\ell+1} e^B = e^B A^{\ell+1}$ holds. Then

$$A^{\ell+1} e^B = A(A^\ell e^B) \stackrel{\text{IH}}{=} A(e^B A^\ell) = (A e^B) A^\ell \stackrel{\text{IH}}{=} e^B A A^\ell = e^B A^{\ell+1}.$$

This concludes the induction proof.

We continue with a proof of $e^A e^B = e^B e^A$. Then

$$\begin{aligned} e^A e^B &= e^A \left(\sum_{k=1}^{\infty} \frac{1}{k!} B^k \right) = e^A I + e^A B + \frac{e^A B^2}{2!} + \dots = I e^A + B e^A + \frac{B^2 e^A}{2!} + \dots \\ &= e^B e^A. \end{aligned}$$

This concludes the proof of item 1.

We continue with item 2.

$$e^{sA} e^{tA} = \left(\sum_{k=0}^{\infty} \frac{1}{k!} (sA)^k \right) \left(\sum_{k=0}^{\infty} \frac{1}{k!} (tA)^k \right) = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{(sA)^k}{k!} \frac{(tA)^{n-k}}{(n-k)!}$$

Dealing with the exponents of A yields

$$e^{sA}e^{tA} = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{s^k t^{n-k}}{k!(n-k)!} A^k A^{n-k} = \sum_{n=0}^{\infty} A^n \sum_{k=0}^n \frac{s^k t^{n-k}}{k!(n-k)!}.$$

Multiplying with $\frac{n!}{n!}$ yields

$$e^{sA}e^{tA} = \sum_{n=0}^{\infty} \frac{A^n}{n!} \sum_{k=0}^n \frac{n!}{k!(n-k)!} s^k t^{n-k} = \sum_{n=0}^{\infty} \frac{A^n}{n!} (s+t)^n = e^{(s+t)A}.$$

For $s = 1$ and $t = -1$, we get $e^A e^{-A} = e^{(1-1)A} = e^{\mathbf{0}} = I$. Therefore, e^{-A} is the inverse of e^A .

Exercise 31

In this exercise, I present a proof of one of Ramanujan's famous equations

$$\sqrt{1 + n\sqrt{1 + (n+1)\sqrt{\cdots}}} = n + 1. \quad (\text{Rn})$$

Here and in the following, a square root means the positive square root. The proof follows M. Sheard's proof in [3] but additional explanations are added. There are some claims for which you should provide a proof.

Let us start by defining for all $1 \leq n \leq m$ the expression

$$g(n, m) = \sqrt{1 + n\sqrt{1 + (n+1)\sqrt{\cdots (m-1)\sqrt{1+m}}}}$$

So $g(m, m) = \sqrt{1+m}$ and $g(n, m) = \sqrt{1 + ng(n+1, m)}$ for $1 \leq n < m$.

Claim 1: For all n, m with $1 \leq n \leq m$, we have $g(n, m) \leq n + 1$.

Prove Claim 1 by a downward induction on n starting from m and proceeding towards 1.

Next we consider the sequence $g(n, m)$ for a fixed n and increasing $m \geq n$.

Claim 2: For all natural numbers $m \geq n$, $g(n, m) \leq g(n, m+1)$ holds.

Prove Claim 2 in a similar way as Claim 1.

When we consider the sequence, then we see that (i) it is bounded and (ii) it is increasing. Recall from your math lecture the following theorem.

Every increasing bounded sequence converges (i.e., has a limit).

Therefore we can, for any positive integer n , define $f(n)$ as follows.

$$f(n) = \lim_{m \rightarrow \infty} g(n, m) = \sqrt{1 + n\sqrt{1 + (n+1)\sqrt{\cdots}}} \leq n + 1$$

We need next that $f(n) \leq f(n+1)$. First we argue that $\forall n f(n) \geq 1$. Suppose there exists an n , for which $f(n) < 1$. Then $f(n) = \sqrt{1 + nf(n+1)} < 1$ which implies that $f(n+1)$ is negative. But this is impossible, because $f(n+1)$ is of the form $\sqrt{1 + (n+1)f(n+2)}$ and we only consider the positive square root.

We argue that $f(n) > n$. Since $g(n, m)$ is an “initial expression” of $f(n)$ and $g(n, m)$ is increasing for increasing m , it suffices to show that $g(n, m) > n$ for a large enough value for m .

Claim 3: Let $m = n^2$. Then for all n , $g(n, n^2) > n$ holds.

Prove Claim 3 in a similar way as Claim 1.

Now we can show that for all positive integers n , $f(n) \leq f(n+1)$ holds. Consider

$$\begin{aligned} f(n)^2 &= 1 + nf(n+1) \\ f(n+1)^2 &= 1 + (n+1)f(n+2) \end{aligned}$$

Now we have

$$\begin{aligned} f(n+1)^2 - f(n)^2 &= 1 + (n+1)f(n+2) - (1 + nf(n+1)) \\ &= (n+1)f(n+2) - nf(n+1) \\ &> (n+1)(n+2) - nf(n+1) \\ &\geq (n+1)(n+2) - n(n+2) \\ &= n+2 > 0. \end{aligned}$$

So, $f(n) < f(n+1)$.

Claim 4: For all n and $k \geq 0$, we have $f(n) > n + 1 - \frac{1}{2^k}$.

Prove Claim 4 by an induction on k .

Note that Claim 4 can be reformulated as follows. First choose an arbitrary n . Then

$$\begin{aligned} \forall k \geq 0 \left(f(n) > n + 1 - \frac{1}{2^k} \right) &\text{ iff } \forall k \geq 0 \left(\frac{1}{2^k} > n + 1 - f(n) \right) \\ &\text{ iff } \forall k \geq 0 \left(|n + 1 - f(n)| < \frac{1}{2^k} \right). \end{aligned}$$

Next we choose an arbitrary $\epsilon > 0$. Instantiate k in such a way that $\frac{1}{2^k} \leq \epsilon$. Then $|n + 1 - f(n)| < \epsilon$ can be derived from $|n + 1 - f(n)| < \frac{1}{2^k}$, $\frac{1}{2^k} \leq \epsilon$ and transitivity. But then we have $\forall \epsilon > 0 |n + 1 - f(n)| < \epsilon$.

Recall that, for $x, y \in \mathbb{R}$, $x = y$ iff $\forall \epsilon > 0 |x - y| < \epsilon$. Therefore, $f(n) = n + 1$.

Solution 31

Claim 1: No solution available.

Claim 2: Choose $m \geq n$ arbitrarily. We perform a backward induction on n starting with $n = m$. Let $\mathcal{P}_m(n)$ denote $g(n, m) \leq g(n, m+1)$.

Base case $n = m$. Then $g(n, n) = \sqrt{1 + n}$. Then

$$g(n, n+1) = \sqrt{1 + ng(n+1, n+1)} = \sqrt{1 + n\sqrt{1 + (n+1)}} \geq g(n, n).$$

Therefore $\mathcal{P}_m(n)$ holds.

Induction hypothesis. Choose $n+1 \leq m$ and suppose $\mathcal{P}_m(n+1)$ is true.

Induction step. We have to show $\mathcal{P}_m(n)$ which is $g(n, m) \leq g(n, m+1)$ true. By induction hypothesis, we have $g(n+1, m) \leq g(n+1, m+1)$. Then

$$g(n, m) = \sqrt{1 + ng(n+1, m)} \leq \sqrt{1 + ng(n+1, m+1)} = g(n, m+1).$$

Claim 3: Let $m = n^2$. Then for all n , $g(n, m) > \sqrt{m} = n$ holds.

We perform a backward induction on n starting with $m = n^2$. Let $\mathcal{P}_m(n)$ denote $g(n, m) = g(n, n^2) > \sqrt{m} = n$.

Base case $n = m$. Then $g(m, m) = g(n^2, n^2) = \sqrt{1 + n^2} > n = \sqrt{m}$. Therefore $\mathcal{P}_m(n)$ holds.

Induction hypothesis. Choose $n+1 \leq m$ and suppose $\mathcal{P}_m(n+1)$ is true.

Induction step. We have to show $\mathcal{P}_m(n)$ true. By induction hypothesis, we have $g(n+1, m) > \sqrt{m} = n$. Then

$$g(n, m) = \sqrt{1 + ng(n+1, m)} > \sqrt{1 + n\sqrt{m}} = \sqrt{1 + n^2} > n.$$

Claim 4: No solution available.

References

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- [2] Mark Joshi. *Proof Patterns*. Springer International Publishing, 2015.
- [3] Michael Sheard. Backwards induction and a formula of Ramanujan. *Mathematics Magazine*, 85(5):364–366, 2012.